Applicant: LUO Xiaoyu
Program: MSc Artificial Intelligence, The Chinese University of Hong Kong

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PERSONAL STATEMENT

The first time I felt the charm of technology was when I assembled my first computer with my own hands. The process of bringing a pile of cold components to life sparked a strong curiosity in me about the principles of hardware-software collaboration. As my studies progressed, I began to focus on deeper questions: how these hardware-driven systems exhibit intelligence under the influence of algorithms and data. During my undergraduate years, diverse project experiences, such as website building, server operation and maintenance, and function development, allowed me to gradually accumulate experience in programming and system construction. They also showed me that current artificial intelligence is becoming the core driver of technological innovation. When I see technologies like AI tools and smart cars turning human imagination into reality, I am even more eager to stand at the forefront of this transformation. I want not only to participate in the design of intelligent systems as a developer but also, from an entrepreneur's perspective, to turn technology into products and services that truly improve people's lives. The combination of AI and entrepreneurship represents the future direction of technological innovation, and it also carries my original aspiration to create the future.

During my undergraduate studies, I systematically built a knowledge system in information technology and AI, and deepened my understanding through the integration of theory and practice. Through courses like *Introduction to Software Engineering*, and *Software System Design and Architecture*, I mastered the overall process of the software development lifecycle by understanding the logic of system architecture design, modular management, and network communication which taught me to analyze problems from a system perspective, design efficient and scalable solutions, and maintain structured thinking in complex projects. In courses closely related to AI, such as *Data Structures and Algorithms*, and *Artificial Intelligence Foundation*, I gradually developed an algorithm-centered way of thinking and understood the entire process from data input and model training to result output where strengthened my abilities in logical reasoning, computational complexity analysis, and algorithm optimization and allowed me to understand the working principles of AI systems from the underlying mechanisms.

Beyond the classroom, I continued to deepen my knowledge through project practice. In a traffic inquiry system developed with classmates, as the front end to develop based on the Vue framework with a completely unfamiliar framework, I read official documents, analyzed open-source projects, and used AI tools for learning and quickly mastered its component-based logic and data interaction mechanisms, and successfully completed the development task. This experience made me deeply appreciate the mutual promotion between theory and practice: theory provided me with ideas to solve problems, while practice allowed me to verify and optimize these ideas in real scenarios. To further consolidate my knowledge, I built my own GitHub repository, which systematically records the source code of projects I participated in, study notes, and development insights that helped me form a sustainable learning system and enabled me to quickly pick up and innovated with new technologies. During a one-month server operation and maintenance internship, I gained a deep understanding of the integrity and complexity of computer systems. My main task involved developing an auto-update script for a bastion host (JumpServer) to facilitate its automatic "drift" to a new server upon failure. Initially, I planned to detect the new environment by checking for mismatched system IDs in configuration files. However, I discovered these files updated automatically during migration, rendering this method ineffective. After multiple attempts and debugging, my solution was to have the script, triggered at system startup, query the cloud provider's API using the machine's ID to obtain its purchase time. If this time was very recent (e.g., within minutes) compared to the script's execution time, it confirmed the environment shift, prompting the script to execute the necessary configuration updates. This process not only successfully ensured the migration but also made me realize that the various modules of a system are deeply interconnected, not isolated. At the same time, when writing backup scripts for the bastion host, I accidentally deleted configuration files due to an operational error. This mistake made me acutely aware of the critical importance of permission management and risk control in system design and operation.

During this experience, my most core gain was the formation of a systematic thinking that considers technological innovation within complex systems. When solving operation and maintenance problems, I not only had to write effective scripts but also design matching work processes and management mechanisms to ensure their stable and efficient operation. This made me realize that a successful AI project is the same. It is not merely about training and optimizing a model, but requires building a complete system that integrates data, algorithms, engineering and business. I deem that my practice of solving systemic problems with limited resources is precisely the prototype of this AI system architecture thinking. I am eager to systematically learn the cutting-edge theories of artificial intelligence in your project and apply this systems thinking to solve larger and more complex real-world challenges.

That is the reason why I am applying for the MSc Artificial Intelligence provided by The Chinese University of Hong Kong. This program's leading position in the field of fundamental theories and ethics of artificial intelligence will provide me with an ideal environment to explore this vision. I look forward to being here and transforming my academic enthusiasm into the ability to address complex challenges at the societal level. The course Artificial Intelligence In Practice captures the most of my attention for its wide coverage in multiple fields, covering data preparation and loading, basic vector operations, multi-layer perception, and convolutional neural networks. I believe that these things will enable me to master the implementation methodology of modern AI systems comprehensively. Among them, I particularly value the practical training on model debugging and performance optimization in the course, as it will allow me to more effectively convert innovative algorithms into verifiable and reproducible solutions in future research. Simultaneously, I also have high expectations for the course Generative Artificial Intelligence, which will direct me to systematically explore the core mechanisms and cutting-edge advancements from diffusion models to generative adversarial networks. What I am eager to deeply understand is their powerful generative capabilities as well as their inherent theoretical bottlenecks and algorithmic limitations.

Technically speaking, I realize that the technical advantage of this project lies in the close integration of its theoretical algorithms and system engineering. It covers core areas such as deep learning, computer vision, and natural language processing, providing in-depth specialized paths in cutting-edge fields such as generative AI, reinforcement learning, and trustworthy machine learning. The top-notch laboratory resources and industrial-level practical projects of his will greatly cultivate my ability to build intelligent systems that are both innovative and practical.

After systematic academic training and diverse practical experiences, I have a clear plan for my future career direction. After graduation, I plan to join an AI technology-driven enterprise, working on AI system development and innovative product design to accumulate comprehensive experience in technology R&D and project management. Within 3 to 5 years, I hope to establish an innovative startup company centered on AI technology, focusing on applying AI to the intelligent solution of real-world problems—such as urban traffic optimization, enterprise information security, and automated system construction.

LUO Xiaoyu

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EDUCATION

Aberdeen Institute of Data Science and Artificial Intelligence, South China Normal University

Bachelor of Engineering in Software Engineering, South China Normal University

Bachelor of Science with Honours in Computing Science, University of Aberdeen ***09/2022-07/2026***

- **GPA**: 3.27/5.0 (82.7/100)
- **Courses: C*omputer Architecture, Java Programming, Web Application Development, Software Testing and Quality, Artificial Intelligence Foundation, Data Mining and Visualization, etc***

INTERNSHIPS

Intern, Beijing Yanling Network Wei Intelligent Technology Co., Ltd. ***07/2025-09/2025***

- Led in composing project planning based on referring to the existing research findings, and collected analysis data to check the feasibility of the codes edited
- Composed a script to extract vulnerability information and analyzed its preciseness and efficiency, aiming at optimizing the script's performance and predicting the overall cost and performance consumption ratio
- Applied LLM to classify and deliver a vulnerability detection to the Web applications and mini-programs, and utilized the taint analysis technique to trace the propagation path of tainted data within the program

Intern, Guangzhou Netlaw Internet Technology Co., Ltd. ***10/2024-12/2024***

Technique Engagement

- Assisted in developing the internal agent management system, and underwent the coding, testing, and optimization of the exception handling module
- Developed functional modules of the system according to project requirements and design schemes, including embedded software development, driver development, bare metal program development, and embedded operating system application development
- Participated in maintaining and upgrading internal software systems, resolving technical issues including automated drift scripting for JumpServer and startup self-check script development

IT Development Assistant Engineer, SmooRe (Shenzhen) International Holdings Limited ***07/2024-08/2024***

- Employed Vue 3 (Composition API) and TypeScript to build the user interface, and optimized the component structure of the page to enhance page loading speed and response efficiency
- Constructed the backend logic to build a Spring Boot microservice architecture using Java, and assisted in designing and implementing API interfaces to support data communication between the front end and the back end
- Integrated databases MySQL into the data storage and management, to promote dynamic pricing, and improve data processing efficiency

PROJECT EXPERIENCES

The Setup and Operation of the Backend Servers and Databases

Programmer, *02/2025-05/2025*

Personal Engagement:

Mastered the technical framework from Java Web to Spring Boot, and built a medium-sized backend service, adopting the classic three-tier layered architecture (presentation layer, business layer, data layer), achieving separation of concerns: the presentation layer handles front-end interactions, the business layer encapsulates core logic, and the data layer manages database operations and effectively enhancing the maintainability and scalability of the code

the Setup and Operation of the Backend Server for the Online Real-Time Weather Query Website

Programmer, *09/2024-01/2025*

Personal Engagement:

Undertook the backend architecture design and core code implementation of the project, and led the completion of the early system modeling (including class diagrams, activity diagrams, etc.), successfully deploying the service to the online environment, and achieving a full-process closed loop from design, development to operation and maintenance

The Front-End Design of the Online Real-Time Bus Query Website

Programmer, *10/2024-12/2024*

Personal Engagement:

Developed the login/search module (using HTML/CSS/JS/Vue), studied the interaction mechanism between the backend API and the database, and participated in UML diagramming, achieving a leap from functional development to system understanding, and possessing the ability to participate throughout the entire process

ADDITIONAL

- Languages: *IELTS* 7.0 (08/2025), Mandarin (Proficient)